

Homework (Carolina Reaper)

- Q1.** A music festival has 3 stages with 5 artists performing on each stage. If the total number of performers is x , what is the unit rate of performers per stage?
(A) 5 performers/stage
(B) 3 performers/stage
(C) 15 performers/stage
(D) 1 performer/stage
(E) 2 performers/stage
- Q2.** A painter uses 2 liters of paint to cover 10 square meters of canvas. What is the unit rate of paint consumption in liters per square meter?
(A) 0.2 liters/m²
(B) 2 liters/m²
(C) 1 liter/m²
(D) 0.5 liters/m²
(E) 5 liters/m²
- Q3.** The frequency of a guitar string is directly proportional to the $\sqrt{\frac{1}{L}}$, where L is the length of the string in meters. If a guitar string of length 0.5 meters has a frequency of 440 Hz, what is the frequency of a string of length 0.25 meters?
(A) 880 Hz
(B) 440 Hz
(C) 220 Hz
(D) 110 Hz
(E) 1760 Hz
- Q4.** A music school has 15 students in a class, and the teacher wants to divide them into groups of 3 for a project. If x represents the number of groups, what is the unit rate of students per group?
(A) 3 students/group
(B) 5 students/group
(C) 15 students/group
(D) 1 student/group
(E) 2 students/group
- Q5.** The cost of renting a studio for a music video shoot is 200 per hour. If a video shoot requires x hours, what is the unit rate of cost per minute?
(A) 3.33
- (B) 5 per minute
(C) 10 per minute
(D) 15 per minute
(E) 20 per minute
- Q6.** A artist uses 12 colors of paint to create a mural on a 4 meter high wall. If the wall is 8 meters long, what is the unit rate of colors per square meter?
(A) 0.375 colors/m²
(B) 3 colors/m²
(C) 1.5 colors/m²
(D) 0.5 colors/m²
(E) 2 colors/m²
- Q7.** The speed of a rhythm pattern is directly proportional to the $\frac{1}{t}$, where t is the time interval in seconds. If a rhythm pattern with a time interval of 2 seconds has a speed of 30 beats per minute, what is the speed of a rhythm pattern with a time interval of 1 second?
(A) 60 beats/minute
(B) 30 beats/minute
(C) 15 beats/minute
(D) 45 beats/minute
(E) 90 beats/minute
- Q8.** A concert has 5 stages with x seats on each stage. If the total number of seats is 2500, what is the unit rate of seats per stage?
(A) 500 seats/stage
(B) 250 seats/stage
(C) 100 seats/stage
(D) 50 seats/stage
(E) 1000 seats/stage

Open Ended Questions (FRQ)

- Q9.** A painter is designing a mural with a pattern that repeats every 2 meters. If the wall is 10 meters long, and the painter uses 3 colors of paint for every 4 meters of the pattern, what is the total number of colors used for the mural? Show your work and explain your reasoning.
- Q10.** A music producer is creating a song with a tempo of 120 beats per minute. If the producer wants to increase the tempo by 10% for a 2-minute instrumental section, what is the new tempo in beats per minute? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to read the questions carefully and identify the unit rate or conversion factor required to solve the problem.
- Tip 2: Remind students to check their calculations and ensure they are using the correct units.
- Tip 3: Emphasize the importance of labeling diagrams and showing work for free-response questions.